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Title: DESIGN GUIDELINE, HIGH PRESSURE DIE CAST AND GRAVITY CAST PERMANENT MOLD	
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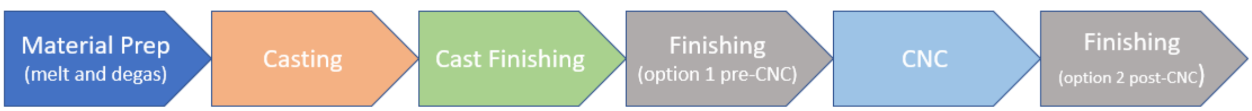
DESIGN GUIDELINE, HIGH PRESSURE DIE CAST AND GRAVITY CAST PERMANENT MOLD

Fabrication process overview

Gravity Fed Permanent Mold (Perm Mold) and High Pressure Die Casting (HPDC or Die Casting) are two metal fabrication techniques commonly used at Amazon Robotics in the production of non-ferrous metal components. At Amazon Robotics, the material typically used is typically aluminum, but can also include magnesium, zinc and other metals. These processes are not suitable for iron, steel, stainless steel. This guideline will consider aluminum as the material for casting. Variances in processing and design consideration for magnesium and zinc should be investigated separately if a particular project has a need.

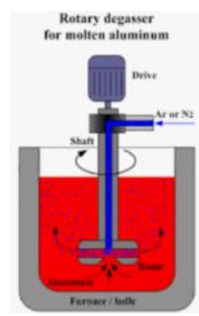
The as-cast part is not suitable for applications at AR without additional machining. Higher precision surfaces and features are created using post-Cast CNC machining on the part (CNC machining to be discussed in a separate design guideline). Parts can be left “raw” or finished with a variety of treatments.

The process flow can be represented as:

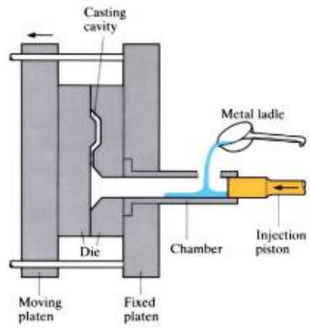


Brief description of each process step

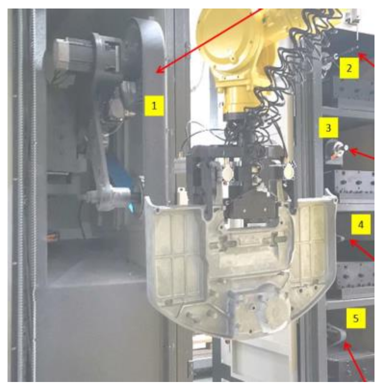
- Material Prep: Aluminum is melted and degassed in preparation for casting. Degassing involves introducing inert gas into the molten aluminum to reduce reaction air and moisture which would allow hydrogen to be absorbed into the aluminum which would turn into trapped bubbles as the melt solidifies.



- Casting: Molten aluminum is introduced into the mold by either gravity (Perm Mold) or via ram-induced pressure (cold chamber high pressure die casting)



- Cast finishing. Gates and vents are removed with trim dies and manual or automated grinding processes.



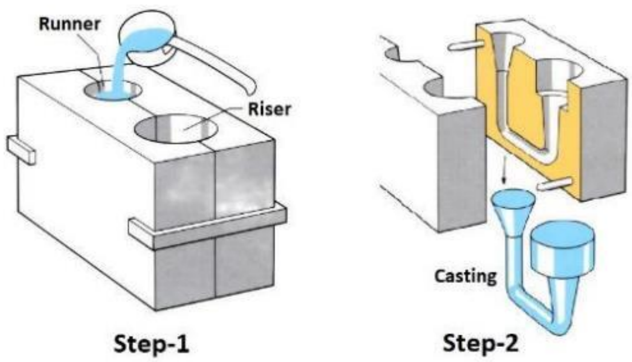
- Finishing (pre or post-CNC): Depending on the requirements for finish (powder coat, anodize, e-coat, chromate, etc), the castings can be finished either before or after CNC

- CNC Machining: Higher precision surfaces and features are created using post-Cast CNC machining on the part. CNC machining design guidelines will be part of a separate document.

Primary Differences Between High Pressure Die Casting and Permanent Mold

In both cases, two-part steel molds[1] are fabricated. The mold halves are held together, then filled with molten metal to create a net shape part, once the part has cooled sufficiently, the mold is opened and the part removed.

- In the case of Permanent mold (also known as gravity casting), the two halves of the mold are painted/coated with ceramic paint, clamped together and molten metal is poured into the mold.



  Gravity pulls the molten metal to the bottom of the mold and the part is filled from the bottom up (hence the term, gravity casting). The ceramic paint is used for thermal management within the mold and causes directional solidification of the molten aluminum as well as improving the flow/fill of the molten aluminum.
- In the case of High Pressure Die Casting, molten metal is loaded into the die casting press and is hydraulically pushed into the mold with an injection piston at high speed and high pressure. The two halves of the mold are hydraulically clamped together. Hydraulic clamping is required to manage the high injection pressure (typically around 8Kpsi). Each square inch of the cast part (including gates, runners, vents) requires 4 tons of clamping pressure. Die casting presses are referenced as to their clamping force in terms of tonnage. Our supply base currently has machines with clamping force up to 4400 tons

[1] While molds are traditionally considered 2-part open and shut with a traditional core (drag) and cavity (cope) side comprising the two halves of the mold, molds are much more complex than this with ejection systems, options for side action created by slides and other mold components.

On past projects AR has had a bias towards die casting to leverage piece part advantage, unless the production demands are low quantity. Permanent mold has traditionally been used on very large, structural parts with thick cross-sections that cannot be achieved with HPDC such as the H-DU front chassis, turntable top/bottom or the S-DU chassis castings.

Process Comparison: High Pressure Die Cast vs Permanent Mold

	HPDC	Perm Mold	Notes
Profile Tolerance	✓		E-W claims as tight as 0.5mm on same side of tool for HPDC, but is part size dependent For reference: HPDC profile on H-DU panhard and tiebards is 0.75mm, on H-DU rear chassis 1.5mm Perm mold on H-DU front chassis and rotary table is 2.6mm, 3.6mm across parting line
Dimensional Tolerance-as cast	✓		Published tables show HPDC at .002mm/mm of length tolerance, Perm Mold at .38mm + .002mm/mm tolerance Across parting line: HPDC additional 0.25mm. Perm Mold additional 0.76 to 1mm. Part size, across parting line or to movable slide all impact tolerance
Part Strength		✓	HPDC cast material is 30-50% stronger than gravity casting equivalents (before HT). With heat treatment, gravity castings can approach but not quite meet a HPDC strength (maybe -5-10%) when considering similar geometery between HPDC and Perm Mold. However given ability to cast much thicker cross sections in Perm Mold, the ability to create a stronger part favors Perm Mold.
Surface finish and Draft	✓		60-120 RMS for HPDC vs 250-420 RMS typical for Perm Mold.
Draft	✓		Required draft is higher for Perm Mold to reduce risk of sticking. 1 to 1.5 degrees for HPDC, 1-3 degrees or more for Perm Mold
Minimum nominal as-cast wall thickness	3-10 mm	4mm and thicker	
Max part size		✓	Max HPDC part size is driven by max 4400 ton press size at our preferred supplier, which equates to ~5500cm^2 maximum projected part surface area at the time of publishing on the larger side of the part. AR uses Perm Mold for parts larger than this
Tooling Cost		✓	All things similar (part size, tooling location-US, Asia, etc) Perm Mold tooling cost and leadtime are less since the tooling is simpler than HPDC Historical tooling costs: HPDC: $15k-300K+, Perm mold: $10K-60K+
Tooling Leadtime		✓	Nominal Leadtimes: HPDC: samples = 12-26 weeks; production = 3-6 weeks after approval Perm mold: samples = 12-16 weeks; production 3-6 weeks after approval
Casting cycle time			HPDC Cycle times are in the 60-100 second range (dependent upon part size) Perm Mold times are in the 240-280 second range (dependent upon part size)
Piece Part Cost	✓		HPDC has faster fill times and lower part production cycle times and is usually more competitive than perm mold
Tool life	✓		We cap tool life for HPDC at 100K shots due to mold degradation (heat checking for instance) We manage perm mold tool life to 60K shots

Applicability of processes based on design stability or development phase

Both High Pressure Die Casting and Permanent Mold are production manufacturing processes. Even lower volume AR-DUs (drive units) in the 10K EAU range will justify the tooling investment to yield a high performing, lower cost part than can be achieved with fabrication techniques suitable for prototyping or lower volume where the tooling investment is not justified. Depending on the part requirements and complexity, a cost tradeoff between perm mold and other low volume cast or CNC processes may show perm mold as cost effect at as low as 1000 pieces annually.

Tooling for permanent mold or HPDC ranges from the $10,000’s upwards of $500,000+ depending on the part size and complexity of trimming and CNC fixtures.

Alternate fabrication processes for prototyping or low volume include:

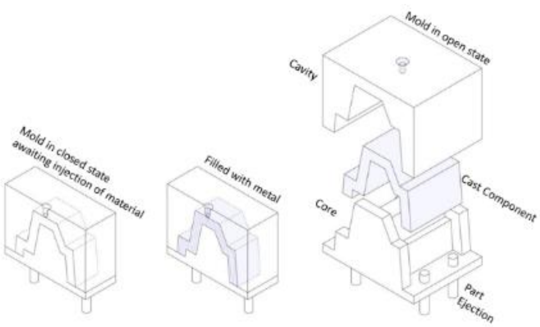
- V-process casting. The closest approximation to the structural performance and finish of an HPDC or perm mold.
- Rubber plaster or plaster molds
- Sand casting
- CNC machining. For low volume or cases where you do not need to have a cast part to replicate as-cast performance, CNC is a suitable alternative for smaller parts with competitive pricing from Asia.

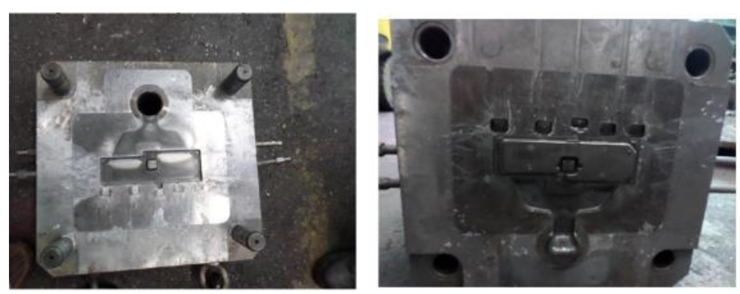
Design for casting guidelines

Key References:
NADCA Product Specification Standards for Die Castings - 11th Edition
Standards for Aluminum Sand and Permanent Mold Castings - 16th Edition 2021

While there are some tolerance and finish limitations with Permanent Mold compared to Die Casting (see table above), the majority of design best practices are common across both manufacturing processes.

Many of the basic design principles of Perm Mold and HPDC are common with injection molding. A two-piece mold is created with a negative space void to be filled with aluminum (or other metal) or plastic in the case of injection mold. The mold is held shut during casting/molding, and then opened once adequate cooling has occurred. The part is then ejected out of the mold.





H-DU ODS Cover Mold in disassembled state

Gates and Vents/Risers

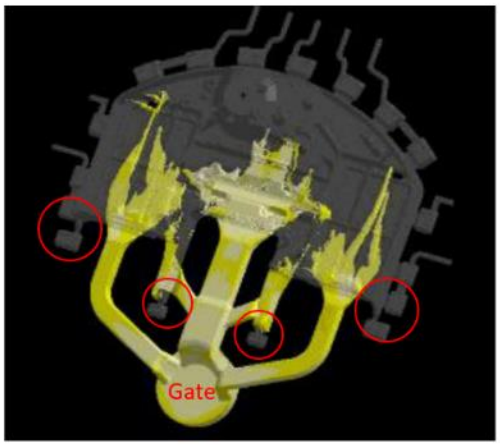
Gates (where the molten metal is introduced into the mold for filling) and vents or risers (where air within the mold is evacuated and excess molten metal is allowed to flow to ensure part fill) are elements of a cast part to consider during part design. Ultimately the supplier will drive the decision on gate and venting, but having basic knowledge of strategy may help through the design process and DFM.

Die casting:

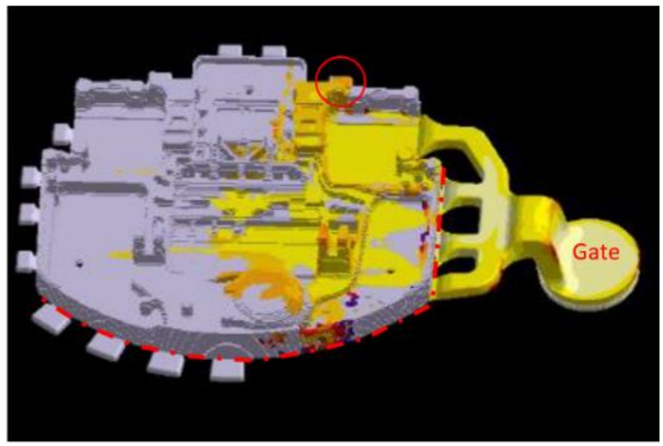
- Gate location will set the general direction of material flow into the part. Material flow direction is usually determined based upon orientation of least resistance to part fill. Impediment to good fill of a part may be changes in part thickness, holes in part, features which sit orthogonal to planned direction of fill.

- Venting is used to promote part fill and “steers” material flow

See comparison below of two similar parts with different gate orientation. Rear chassis flow will be more direct along the axis of the hinge arms, while the Front chassis flow is better across the part due to the general orientation of ribbing and pockets in the part. In both designs, venting is added to ensure good fill in proximity to the gates (see red circles). Also notice the location of the parting line in red-dashed line and how gating and venting are along the parting line edge of the part.



Rear Chassis



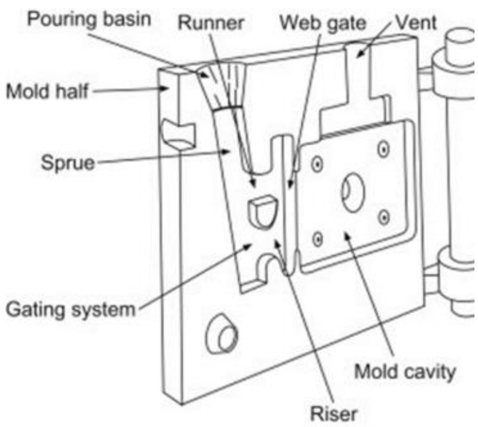
Front Chassis

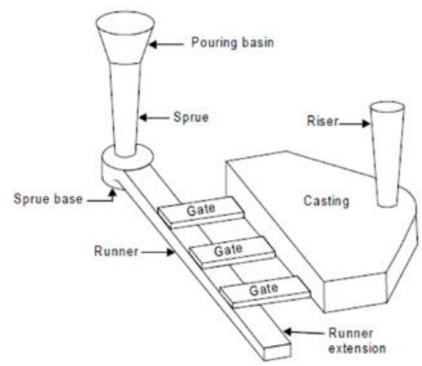
The basic principles of gating and part orientation relative to gates are similar for both die cast and perm mold. Impediments to flow include ribbing orthogonal to flow direction, large holes which divert flow paths. Orthogonal ribbing flow disruption can be mitigated by feeder ribs added in the direction of flow if the design can accommodate. In the case of holes impeding flow, material bridges can be added and machined out alter.

Perm Mold: Due to the gravity fed nature of a permanent mold part with molten metal being poured in rather than rammed in under pressure, gate and riser schemes for permanent molds are typically larger in geometry with a smaller number of risers than comparable die cast vents.

- Gate system: Part orientation and gate location optimized to eliminate the risk of trapped air in a gravity fed system

- Vents/Risers: Allow air to escape the mold cavity and provide overflow to ensure part fill. Simplified views of a gravity fed systems shown below.



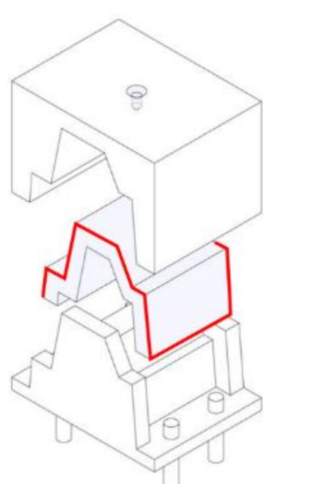


Parting Lines

The parting line is the perimeter on the casting which is the separation point of the two halves of the die casting mold.

The parting line defines the draft plane in the part, so needs to be carefully considered during part design.

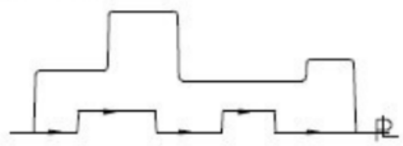
In the example, it follows the red line shown.



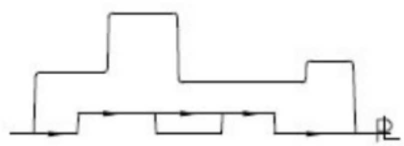
While complex parting lines may be required to accommodate the product design, there are advantages to keeping parting lines as simple as possible, with options considered to trade off between part design and mold complexity.

- Simplified tooling (cost and leadtime)
- Since gates and vents are required for casting and are at the parting line, trim dies will be easier to design and fabricate on a part with a simple parting line than a complex one.
- Tool life, fit and finish. It is easier to maintain a planar parting line than a complex one.

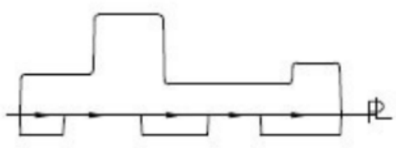
The following example shows decision making tradeoffs between part design, draft and parting line complexity.



Initial design. Follows edge of part
More complex fit between core and cavity sides of the mold



Change proposal #1: reduces complexity of the mold, but requires draft changes on part.

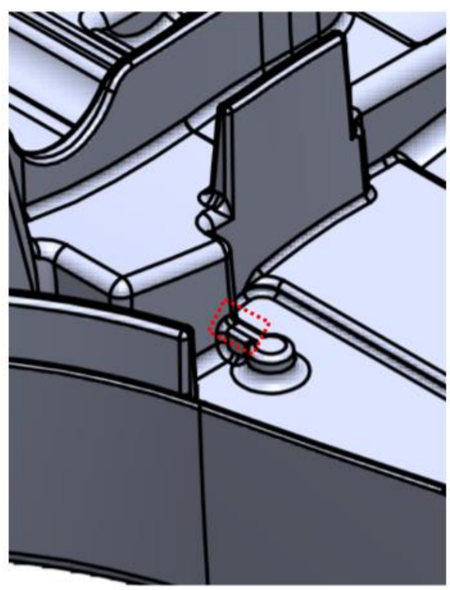


Change proposal #2: Planar parting line-easiest from a tooling perspective, but requires additional draft changes on part.

Consistent/Uniform Wall Thickness, Promoting Good Part Fill.

Maintaining consistent wall thickness promotes good material flow and part fill. This is critical in die casting. Permanent molding using a gravity feed enables additional mold fill as material starts to cool
Where thickness changes must occur, avoid abrupt thickness transitions. Abrupt changes create material turbulence and pressure changes which can result in porosity.
Thick sections are also susceptible to porosity in die castings since air bubbles can form in thick sections as material cools/contracts. This is referred to as “shrink porosity”.

Hanging/isolated features. Hanging features are considered part features which are freestanding, yet adjacent to other features. When possible, these features should be bridged with material to those adjacent features to promote material flow. Other reasons for bridging would be to eliminate areas of thin mold steel. See the *Minimum Feature Size* section below



Bridge features adjacent to side walls to promote material flow
Also note extensive use of inside and edge radii

Corner Radii and Fillets

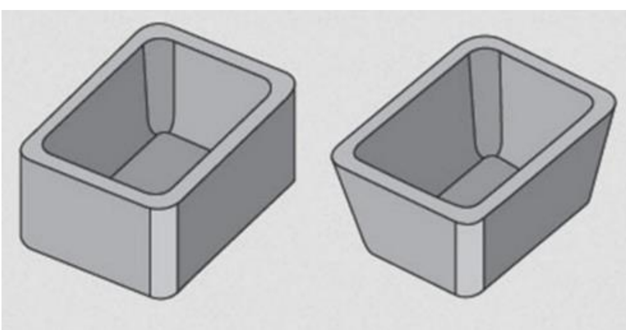
Both internal and external corners in a cast part should have radii added.
Minimum inside R of 0.5mm is a good general practice on die cast parts. If this cannot be achieved, any radius is better than leaving a sharp corner in the product design.
External radii to match the inside R + wall thickness. This helps to maintain uniform wall thickness.
Reasons for adding radii include:

- Sharp corners in the mold that create inside R in a part are impossible to maintain over the life of the mold.
- Sharp inside corners in both the part and the mold are stress concentrators.
- Sharp inside radii

Exceptions:
Edges at the parting line (or other places where different parts of the mold meet) shall not have radii unless the function of the component requires a radius (ie-sharp edge that will be handled, etc.). In this case, a detailed DFM review is required to make sure mold design best practices are followed.

Draft

In general, vertical walls cannot be left vertical in the axis of the mold opening/closing. Molten metal and plastic shrink when they cool, which results in the part sticking against the core side of the tool. In addition, surface roughness creates additional points of friction between the cast part and the core and cavity of the mold. Without draft, a vertical wall on the cast component and the vertical wall of the mold will have significant friction to overcome when trying to remove the cast part from the mold (the process of part removal from a mold is called ejection). This friction can cause part warpage or breakage upon ejection and can result in a cast part being stuck on the core. Adding draft to the part reduces the amount cast part to mold interface. As the part starts to eject from the mold, the drafted surfaces on the mold and the cast part immediately separate.



Examples of a part without draft (left) and with draft (right)

Draft is critical to consider during part design since part geometry can be affected in ways that effect wall thickness, or interfaces to mating parts.
General draft guidelines:

- Die cast parts: Assume 1.5 degrees minimum draft. There may be cases where more or less draft may be required and can be determined during a supplier DFM review.

- Perm mold: Assume 3.0 degrees minimum draft. There may be cases where more or less draft may be required and can be determined during a supplier DFM review.

Note: For walls that need to be truly vertical, consider post-CNC machining on interior walls and/or adding slides/side action on exterior surfaces.

Undercuts and 0 degree draft conditions.

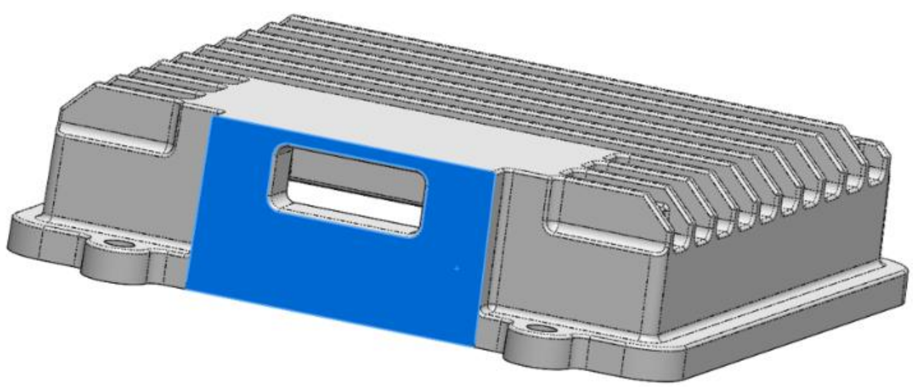
Undercuts in die casting are part features that cannot be manufactured with a simple two-part mold, because material is in the way while the mold opens or during ejection.

0 degree draft refers to the state outlined in the "draft" section above where a wall in the direction of mold open/close is not drafted to enable part ejection

Managing undercuts

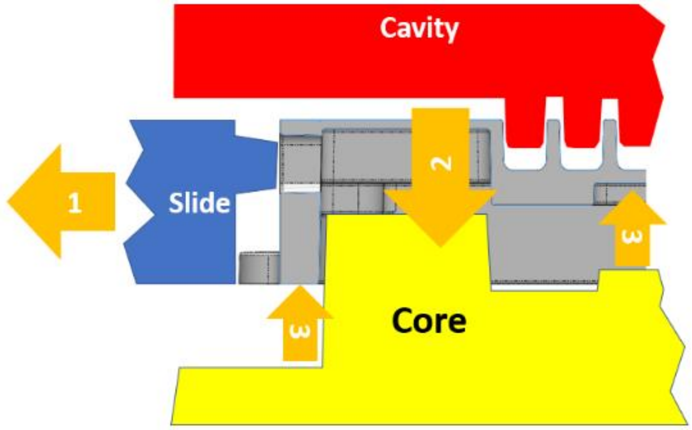
- Side action/slides: These are additional pieces of tool steel used to create the undercut and are moved out of the way either automatically or manually. There is a tooling cost increase to add a slide (10% of tool cost is a good budgetary value)
- Post-casting CNC. These features can be machined in after casting. The additional cost of CNC can be quantified to compare to the additional mold cost for a slide.
- Other options: Ejection Lifter or Pass core. These are not common with our primary supply base for die casting. Consult your supplier for more feedback on feasibility.

An example of an AR part with both undercut and 0 draft conditions is 420-06155 ENCLOSURE, ARIMA, H-DU

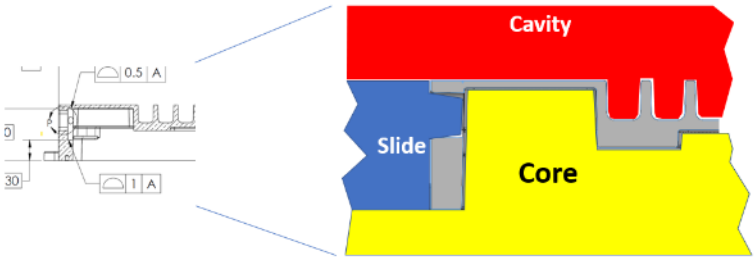


The blue face and rectangular hole are created using a slide on 420-06155
Blue face= 0 draft
Connector cutout = undercut

Basic mold construction may be an aid to visualize how side action is implemented in a mold



Basic Mold Movement including a slide



Tool Construction of the Slide to make the connector opening on 420-06155
Mold in closed position

Tool movement includes

1. The slide moves out of the way to eliminate the undercut situation. Depending upon the mechanism employed to create movement, it may occur concurrently with #2 below.
2. Core (with cast part) moves away from the cavity side of the mold.
3. The cast part is ejected from the core side of the mold.

Side action/slide considerations.

- Slides impact tooling cost and leadtime -avoid if possible
- Tolerance: When considering tolerance, use tolerances 'across parting line" or "from fixed steel to movable steel". There is additional tolerance to consider. Post-cast CNC may be required to achieve the needed accuracy.
- Consider side action and CNC processes together. While side action in a tool can eliminate post machining of casting, an analysis should always be completed to justify using a slide instead of using CNC operations to achieve the desired geometry.

Cast part Geometry consideration for CNC

As noted, post-cast CNC machining of cast parts is common to achieve tighter tolerances, smoother finishes or in areas where the needed features cannot be cast.
Some things to factor into as-cast design:

Adding material for CNC machining. For cast surfaces that need to be machined for precision or finish reasons, additional "over-cast" material must be added to ensure that there is sufficient cast material to machine to the required dimension. This extra metal is commonly referred to as "machining stock".

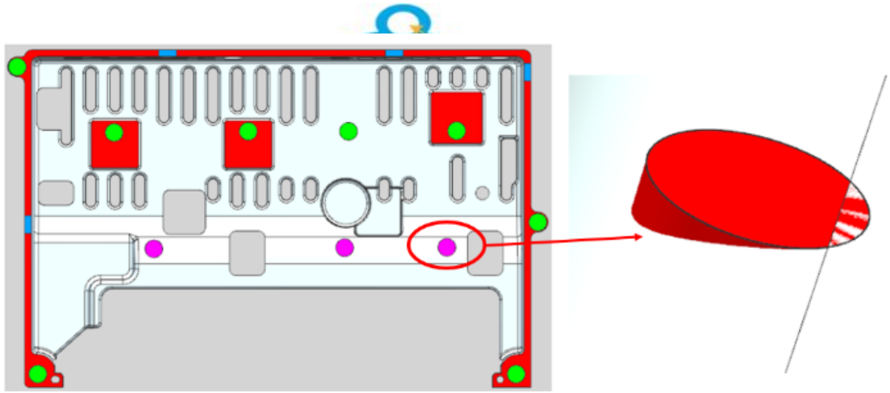
- For die casting: add 1mm of over-cast material. Do not add excess material since the deeper a part is machined from the skin, the greater risk of hitting porosity.
- For perm mold. add 1.5-2mm of over-cast material. Excess material is less risky with perm mold due to reduced risk of porosity, however, excess material will impact CNC time.

Air cuts. To mitigate risk of profile tolerance issues in critical areas due to tool wear which will increase surface size, an "air cut" can be specified that will include a CNC cutter path at maximum material condition.

Other Cast Feature Design Considerations

- Minimum wall thickness
  - Die cast:
    - Nominal 3mm (but thicker preferred)
    - Minimum feature thickness: 2.0mm
  - Perm Mold
    - Nominal 4 mm (but thicker preferred)
    - Minimum feature thickness: 3-4 mm with supplier feedback
- Minimum mold steel thickness (feature to feature gap)
  - Die cast: 3mm (but thicker preferred)
  - Perm Mold: 6mm if height of mold is 20 mm, but can go thinner if mold feature is shallower. Respect ~1:3 tool steel width:height ratio
- Resultant approximate fin pitch (factor in draft) when designing heat sinks
  - Die cast: 2.5mm fin top + 3mm tool steel gap + 2 degrees draft (rib height*sin2 degrees)*2 ribs (note:it is possible to have more aggressive fin pitch with a tradeoff in tool life if tall, narrowly spaced fins are required, with concepts up to 80mm tall, 1.6mm wide at top, 1.5 degrees draft having been explored)
  - Perm Mold: 3mm fin top + 4mm tool steel gap + 2 degrees draft (rib height*sin2 degrees)*2 ribs
- Minimum as-cast hole diameter.
  - Die cast: For holes 5mm DIA or larger, cast the hole. For holes <5mm DIA, they should be created with post- cast CNC. For finished holes right at the 5mm DIA boundary, review with your supplier since profile tolerance needs to be considered.
  - Perm Mold: Minimum as-cast hole diameter driven by maintaining adequate draw and draft angle as well a flow path of pour. As-cast hole diameter depends on final hole depth. Consult supplier for proper blind hole diameter and depth ratio.
- Unusual part geometry. Highlight parts for early DFM that have flat sections without ribbing, parts with large asymmetry, etc. These may have a larger propensity for warp.

- Ejector pin pads. For molds using ejection systems (all die cast molds), there may be areas where circular flat pads need to be added to the part to enable part ejection. These are often prevalent on parts which do not have large surface area orthogonal to direction of ejection



Ejector pin locations shown in magenta sit on an angled surface. Die cast DFM may result in a request to add a flat pad if component clearance allows, or, to determine alternate ejection plans

- Datum pads. Cast datums should be established on the same side of the die half, either cavity side or core side. These as-cast datums are used for as-cast inspection and for any initial CNC operations which are based off of them (including establishing as-machined datums). As the mold tool begins to wear, the variability across the parting line will grow. Therefore, keeping the cast datums on one side will prevent the datum reference frame from varying over time. Choose the die half with the most functional requirements or mating features. Avoid selecting cast datums features that are subject to variability like ejector pins, areas near parting lines, vents, and gates. Select cast datum features that will not become machined in order to maintain traceability of datums from cast to machined. As part datum strategy is developed, consider adding flat pads if possible on surfaces that are not orthogonal to datum structure. This works well on surfaces orthogonal in the direction of mold open/close. To maintain the ability to re-inspect a finished part, cast datums should always be kept on the part itself, not on tabs that will be machined off in later processes.

Materials and Finishes

Materials

The primary alloys of aluminum we use for casting include:

Die Casting:

- A380. General purpose alloy used in the majority of die castings at AR
- AlSi12(Fe). A newer (to AR) alloy with improved thermal conductivity properties. ~ 30% better thermal conductivity than A380 with a minor sacrifice in material properties

Permanent Mold: The primary alloy used on our permanent molded parts is 319.0

Finishes

There are a variety of finishes which can be utilized for both decorative and functional applications

Finish	Description	Cosmetics	Cost	Electrically Conductive?	Corrosion Protection?	Apply Pre-CNC machine?	Color options?	Notes
Raw Casting	As-cast part	As-cast. Not great	None	Yes-but risk of oxidation	No	N/A	N/A	
Chromate Conversion	Corrosion protection. Can be electrically conductive depending on the level of corrosion	Not used for cosmetic reasons	$	Can be applied to be conductive or as an insulator	Yes	Yes if CNC'd surfaces do not need finish, but typically post CNC since finish is clear.	Clear is the most common	Wet/Dip process most often applied after CNC
E-coat	Cosmetic or corrosion protection	WIl provide a semi-uniform surface. Thinness of finish will show as-cast surface quality.	$	No	Yes	Yes if CNC'd surfaces do not need finish	Black with little control with our current supply base over the color specs	Wet/Dip process most often applied after CNC since masking for a dip process is difficult. Finish is thin so thermal impedance is minimal. Wet process is difficult to mask. No experience at AR with applying before CNC and risk of scratching the finish.
Anodize	Cosmetic and Corrosion protection	Can be applied heavier than e-coat to mask as-cast quality	$$	Not usually	Yes	Yes if CNC'd surfaces do not need finish	Multiple, but mroe $$$	Wet/Dip process most often applied after CNC since masking for a dip process is difficult. Finish is thin so thermal impedance is minimal. Wet process is difficult to mask. No experience at AR with applying before CNC and risk of scratching the finish
Powder Coat	Cosmetic and Corrosion protection	Very good cosmetics	$$	No	Yes	Yes if CNC'd surfaces do not need finish	Multiple	Dry powder process. Prototypes to date have shown that powder coat is robust enough to be applied before CNC machining

Process capabilities

As noted previously, die casting will yield a tighter tolerance as-cast part than the permanent mold process can achieve. If the part design can accommodate greater as-cast profile, linear and across parting line tolerance, precision can still be achieved where needed via CNC machining.

While our supply base for high pressure die cast and permanent mold will provide tolerances based upon their capabilities which may be tighter than industry standards. However, if you can design to industry association published tolerances, you can create parts that are manufacturable regardless of supplier selection.

Example tolerances for HPDC and Perm Mold are included below. For a comprehensive tolerancing guide, is better to refer to industry standard data.

- Die Casting: The North American Die Casting Association (NADCA). *NADCA Product Specification Standards for Die Castings* 11th Edition 2021
- Permanent Mold: The Aluminum Association. Tolerances published as part of *STANDARDS FOR ALUMINUM SAND & PERMANENT MOLD CASTINGS* 16th Edition 2021; Engineering Series

General things to consider regarding tolerances:

- Linear tolerance is almost always considered as a % of unit length per unit length. For instance: Current NADCA dimensional tolerance is +/-0.25mm per 25mm, + additional 0.025mm per each full 25mm
- Tolerances across the parting line are always greater than on a single side of the tool. Projected part area may impact tolerance (more area=greater tolerance across parting line due to greater pressure)
- Tolerance to a moving core (slide or other side action): Similar to parting line tolerances, there is additional tolerance when considering dimensions that span from fixed steel to a moving core

Linear tolerance on one side of a mold example:

		Perm Mold	Die Casting Standard Tolerance	Die Casting Precision Tolerance
Dimensions on one side of mold	Basic Tolerance up to 25mm	± 0.38mm	± 0.25mm	± 0.05
	Additional Tolerance for each additional 25mm	± 0.051mm	± 0.025mm	± 0.025
Note regarding Standard vs Precision tolerances Normally, the higher the precision the more it costs to manufacture the part because die wear will affect more precise parts sooner. Production runs will be shorter to allow for increased die maintenance. Therefore the objective is to have as much tolerance as possible without affecting form, fit and function of the part.				

The geometry, feature and mold design cases for which tolerances are considered in the NADCA and Aluminum Org guidelines are extensive:
NADCA:

- Linear Dimensions Tolerances
- Parting Line Tolerances
- Moving Die Component Tolerances
- Angularity Tolerances
- Concentricity Tolerances
- Parting Line Shift
- Draft Tolerances
- Flatness Tolerances

Permanent Mold:

- Linear Dimensions Tolerances
- Parting Line Tolerances
- Side Core Tolerances
- Flatness Tolerances
- Profile Tolerances

Secondary operations

The majority of secondary operations which occur on castings is related to either part finishing (already discussed) or CNC machining where required for precision surfaces. CNC machining is an in-depth topic that will be covered in its own design guideline.
Other secondary operations on castings that are common at Amazon Robotics include:

- Tapping threads-completed as part of the CNC operation. Note that use of thread forming fasteners is an option that could eliminate pre-tapped holes to reduce cost.
- Press fit pins. We rely on our casting suppliers to press fit pins into place. The pin is included on the BOM and we use standard recommendations for hole diameter to specify in our data, but allow our suppliers latitude to CNC the dowel pin hole to their recommended diameter based upon their experiences

Cost drivers

Tooling costs. The following factors impact tooling cost and leadtime.

- As previously noted, High Pressure Die Cast Tooling will have higher costs and longer leadtime than a comparable mold for Gravity Cast/Permanent Mold.
- Mold and trim die complexity.
- Complex parting line, side action or slides-increased tooling cost and ongoing maintenance costs.
- Machining setups. Each machining setup will require a unique fixture.

Piece part cost. The following factors impact the piece part cost and leadtime.

- Material: There may be differences in material cost for various alloys. This can be reviewed with a supplier. At the time of publishing the cost delta between die cast alloys A380 and AlSiFe(12) was negligible.
- Cycle time. As previously noted, High Pressure Die Cast Tooling will have lower as-cast piece part costs than a comparable part produced via Gravity Cast/Permanent Mold, due to shorter cycle times for die casting.
- Automation. Die cast presses and processes lend themselves to greater levels of automation which reduce labor costs.
- Part Size. Larger parts require more raw material. They may also have slower cycle times, which decreases machine utilization.
- Machining operations and total machining time.
- Finishing. See options above for relative cost impact. The finishing process selected plus any masking requirements will impact cost.